Independent ordinary Kummer fiber review

Ordinary Kummer fiber: reviewed conditional derivation

Shared S=b1² → at most two candidates, differing by simultaneous negation

S=0 requires R1=0 for validity; repeated support does not alter polynomial reduction

Neighbor fiber property + 2G≠0 → at most one distinct signed match

Existence still needs candidate completeness, chart identity and successful projections

11 producer + 7 reviewer fixed assertions; written universal argument, not Lean proof

Matrix-square arithmetic is separate from a geometric two-torsion translation law

Result. Validated at the stated scope as a written derivation. This is not a universal Lean theorem or verification of all native coordinate formulas.

Fiber and zero square. Reducing $(b_0 + b_1X)^2$ modulo $X^2 + a_1X + a_0$ yields the stated two equations without requiring squarefree support. Assuming the shared square identity and a valid divisor, nonzero S fixes b_1 up to sign and then b_0 uniquely in characteristic other than2. For $S = 0$, $b_1 = 0$ and validity requires $R_1 = 0$; the source generates candidates before applying this validity filter. Global deduplication limits the generated vector to two distinct candidates even though the zero branch is visited twice.

Neighbor and geometry. The neighbor argument proves uniqueness under the quotient-fiber property on the neighbor domain and $2G \neq 0$. The alternative $2D = 0$ identifies the two candidates. Existence separately needs coordinate correctness, candidate completeness and successful projections. Ordinary-chart algebra does not cover boundary neighbor charts automatically. Matrix-square arithmetic proves invertibility, not translation by a specified two-torsion point.

Native controls. Eleven producer assertions and seven independent assertions passed fresh Rust runs. Reviewer checks include inconsistent $S = 0$ residual, repeated support and characteristic2 denominator. These fixed checks are illustrative, not universal geometric evidence. One initial build had ambiguous integer literal types; failed source/stderr and absent-binary invocation remain. Explicit i64 literals and fail-closed replay corrected the setup.

Custody and scope. Source is pinned to publication125c631e4005785e344167d8f699b4bf23415fe6. Prior graph/polynomial familiarity is disclosed, without a blind-review claim. See REVIEW.md, status.json, receipt.json and replay.sh. No enumeration, curve computation, timing, scientific execution, repository edit, cloud operation or Python research occurred.